

TP2 :Services de Réseaux

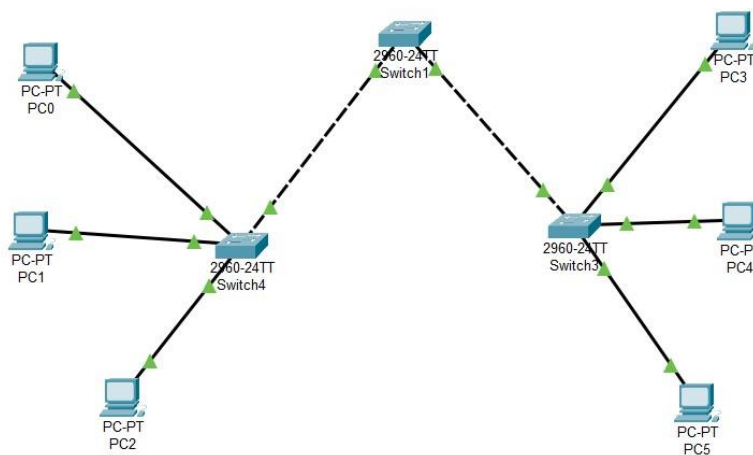
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TD1-TP1

L2INFO

Objectif: Configurer et vérifier le fonctionnement d'un réseau local virtuel (VLAN) . Ce TP nous permet de manipuler les VLAN : leur création, leur attribution aux ports et leur vérification.

Partie 1 : afficher la configuration de VLAN par défaut



-> la topologie du réseau et les liens entre les équipements.

Configuration des adresses IP :

Configuration des interfaces des hôtes avec les adresses IP, les masques de sous-réseau et les passerelles par défaut indiqués dans la table d'adressage comme suit :

PC0

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0002.4A81.7AA6

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 172.17.10.21

Subnet Mask 255.255.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::202:4AFF:FE81:7AA6

PC1

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0030.A375.8A57

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 172.17.20.22

Subnet Mask 255.255.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::230:A3FF:FE75:8A57

PC2

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 000D BD79 5BD7

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 172.17.30.23

Subnet Mask 255.255.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::20D:BDFF:FE79:5BD7

PC3

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00D0 9706 0364

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 172.17.10.24

Subnet Mask 255.255.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::2D0:97FF:FE06:364

PC4

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00D0 FF32 DD05

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 172.17.20.25

Subnet Mask 255.255.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::2D0:FFFF:FE32:DD05

PC5

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0060 2F16 377D

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 172.17.30.26

Subnet Mask 255.255.0.0

IPv6 Configuration

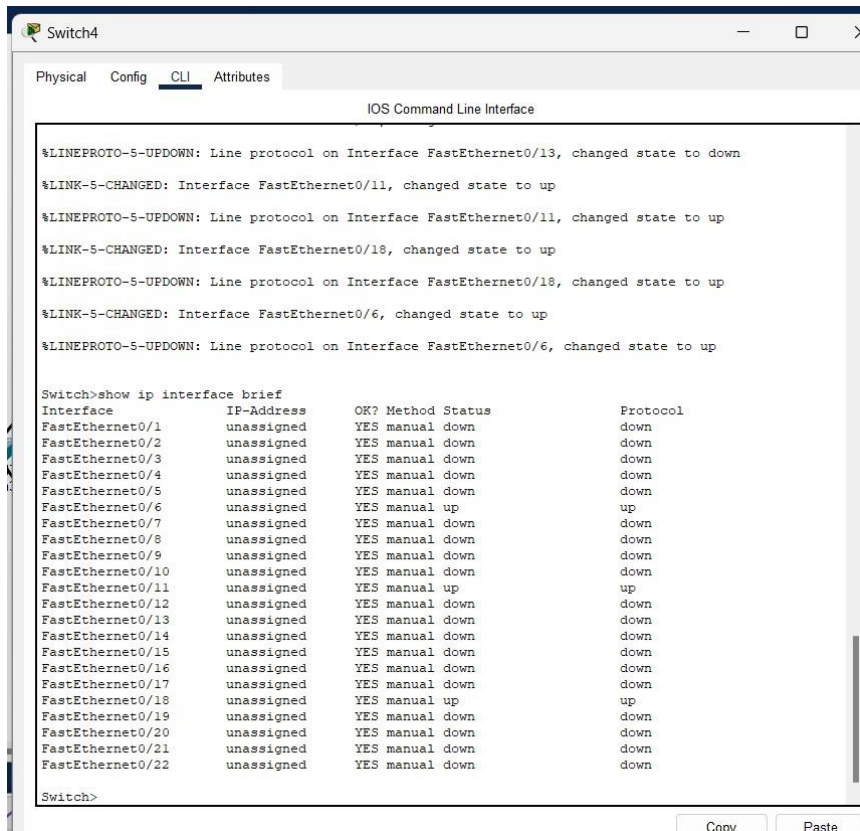
☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::260:2FFF:FE16:377D

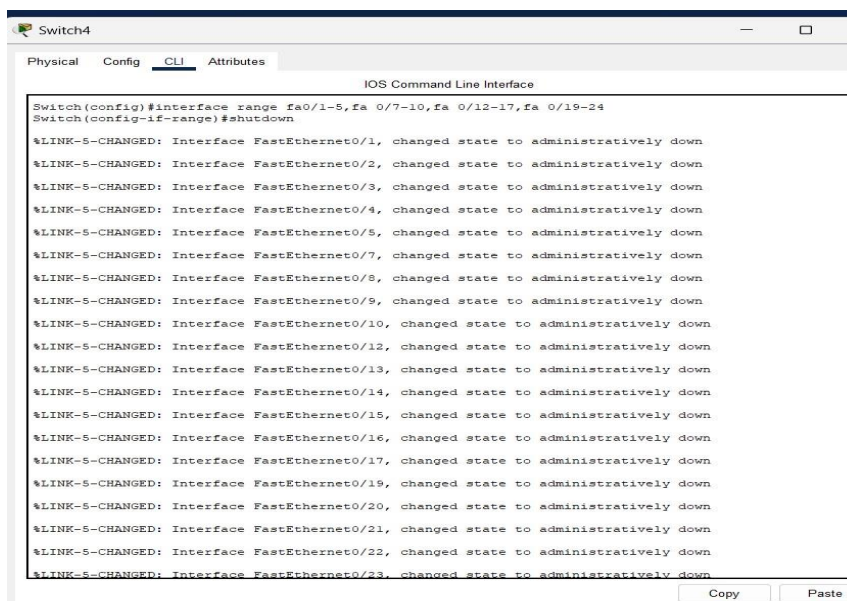
Affichage des interfaces :



The screenshot shows a network switch CLI window titled "Switch4". The "CLI" tab is selected. The window displays the output of the command "show ip interface brief". The output is a table with columns: Interface, IP-Address, OK?, Method, Status, and Protocol. The table lists 22 interfaces (FastEthernet0/1 to FastEthernet0/22). The status of each interface is either "down" or "up".

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	down	down
FastEthernet0/2	unassigned	YES	manual	down	down
FastEthernet0/3	unassigned	YES	manual	down	down
FastEthernet0/4	unassigned	YES	manual	down	down
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	up	up
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	up	up
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	up	up
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down
FastEthernet0/22	unassigned	YES	manual	down	down

Shutdown des interfaces libres dans switch 1 et 2 et 3 :



The screenshot shows a network switch CLI window titled "Switch4". The "CLI" tab is selected. The window displays the output of the command "interface range fa0/1-5,fa 0/7-10,fa 0/12-17,fa 0/19-24" followed by "shutdown". The output shows that all interfaces in the specified range have been successfully shut down.

```
Switch(config)#interface range fa0/1-5,fa 0/7-10,fa 0/12-17,fa 0/19-24
Switch(config-if-range)#shutdown
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down
```

Switch3

Physical Config CLI Attributes

IOS Command Line Interface

Switch(config)#interface range fa0/1-5, fa0/7-10, fa0/12-17, fa0/19-24
Switch(config-if-range)#shutdown

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down

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Switch1

Physical Config CLI Attributes

IOS Command Line Interface

Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fa0/1-24
Switch(config-if-range)#shutdown

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to administratively down

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- Affichage des VLAN actuels :

Sur le switch S1, on a tapé la commande **show vlan brief** pour lister tous les VLAN configurés par défaut. On remarque que toutes les interfaces sont associées au vlan 1.

```
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
```

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- Vérification de la connectivité entre les PC d'un même réseau :

Ici, si les pings réussissent, les PC du même VLAN peuvent communiquer. Si non, il y a une mauvaise configuration.

- L'envoi de ping de pc2 vers pc5

```
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.17.30.26

Pinging 172.17.30.26 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.17.30.26:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

☐ Top

- L'envoi de ping de pc0 vers pc4 :

```
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.17.30.23

Pinging 172.17.30.23 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.17.30.23:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

->Ils ne peuvent pas se connecter car ils ne sont pas sous le même sous-réseau.

Partie 2 : configuration des VLAN

- Créer et nommer des vlan sur S1 :

On a utilise la commande **vlan <numéro>** pour créer les VLAN (10, 20, 30, 99) et attribuez-leur un nom avec **name <nom>**.

```

Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name faculty
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name students
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name managment
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name guest
Switch(config-vlan)#

```

- **Vérification de la configuration VLAN :**

On tape la commande show vlan brief :

```

S2-S3-CONFIG_1: Configured from console by console
Switch#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	faculty	active	
20	students	active	
30	guest	active	
99	managment	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

Switch#

```

- **Créer les vlan sur S2 et S3 :**

On tape les mêmes commandes utilisée précédemment :

```

Switch>
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name faculty
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name students
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name managment
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name guest
Switch(config-vlan)#

```

```

Switch>enable
Switch#vlan 10
      ^
% Invalid input detected at '^' marker.

Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name faculty
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name students
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name guest
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name managment
Switch(config-vlan)#

```

Partie 3 : attribution des vlans aux ports

• Attribuer des vlans aux ports actifs sur S2 :

On doit maintenant associe chaque interface au vlan qui lui correspond comme suit :

```

Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface f0/11
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface f0/18
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#interface f0/6
Switch(config-if)#switchport mode
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#

```

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. Attribution des vlans aux ports actifs sur S3 :

```

Switch(config)#interface f0/11
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface f0/18
Switch(config-if)#switchport access vlan 20
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#interface f0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#

```

- **Vérification qu'il n'y a pas de perte de connectivité :**

->Comme on peut voir dans cette étape la connectivité est effectuée avec succès.

```
C:\>ping 172.17.10.24

Pinging 172.17.10.24 with 32 bytes of data:

Reply from 172.17.10.24: bytes=32 time<1ms TTL=128
Reply from 172.17.10.24: bytes=32 time<1ms TTL=128
Reply from 172.17.10.24: bytes=32 time<1ms TTL=128
Reply from 172.17.10.24: bytes=32 time<1ms TTL=128

Ping statistics for 172.17.10.24:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.17.30.26

Pinging 172.17.30.26 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.17.30.26:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

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- **Configurer le mode trunk :**

On doit passer par les étapes suivantes :

```
>Switch(config-vlan)#name management
Switch(config-vlan)#interface range G 0/1, G0/2
Switch(config-if-range)#switchport mode dynamic desirable

Switch(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
exit
Switch(config)#
```

->Finalement on doit effectuer le test de ping :

```
C:\>ping 172.17.10.24

Pinging 172.17.10.24 with 32 bytes of data:

Reply from 172.17.10.24: bytes=32 time<1ms TTL=128
Reply from 172.17.10.24: bytes=32 time<1ms TTL=128
Reply from 172.17.10.24: bytes=32 time<1ms TTL=128
Reply from 172.17.10.24: bytes=32 time<1ms TTL=128

Ping statistics for 172.17.10.24:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.17.30.26

Pinging 172.17.30.26 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.17.30.26:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Top

Conclusion :

Cet exercice couvre les étapes clés pour configurer et gérer les VLAN sur un réseau. Une fois que la configuration par défaut des VLAN est vérifiée, les switchs sont configurés pour créer et nommer des VLAN pour découper le réseau en groupes indépendants et logiques. Les ports des switchs sont ensuite affectés au VLAN approprié pour entrer en contact avec d'autres dispositifs du même réseau virtuel. Cette configuration organise, sécurise et gère également le trafic via le réseau.

